

AIA Project Briefs

Automatic Image Analysis – TU Berlin

Summer Semester 2026

These project briefs describe five semester topics. Each topic asks you to compare a classical image-analysis pipeline with a modern representation, evaluate both fairly, and explain where each approach works or fails.

Topic	Name of topic
Topic A	Shape and Contour: Leaf Species Recognition
Topic B	Bayesian Decision and Uncertainty: Skin Lesion Triage
Topic C	Scale-Space and Invariant Detection: Matching Across Viewpoints
Topic D	Self-Supervised Representation Learning: STL-10 Linear Evaluation
Topic E	Video Understanding: Temporal Action Recognition on UCF-101

How to read this document

Start from the dataset and task, then check the classical and modern components. The expected outputs tell you what should appear in the presentation and final report.

Project Topics

Topic A – Shape and Contour: Leaf Species Recognition

Dataset	Flavia Leaf Dataset: 1,907 leaf images, 32 plant species. Alternative: Leafsnap.
Core task	Given a segmented leaf silhouette, classify the plant species.
Main question	How far can shape alone go, and when do learned image features add useful information?

Required work

- Classical pipeline: Fourier descriptors of the normalized boundary and Hu moments.
- Classifiers: naive Bayes and k -NN on a stratified train/test split.
- Modern pipeline: frozen MobileNetV2 features with a linear SVM.
- Compare all methods on the same split.

Expected outputs: confusion matrices, per-class accuracy, example leaves that are easy or hard, and a short discussion of when boundary information is sufficient.

Theory focus: Fourier descriptor invariance, Hu moments, and the independence assumption in naive Bayes.

Topic B – Bayesian Decision and Uncertainty: Skin Lesion Triage

Dataset	HAM10000: dermoscopy images with diagnostic labels. Related challenge: ISIC 2018.
Core task	Binary melanoma-vs-benign triage with calibrated uncertainty.
Main question	How should a classifier behave when false negatives are much more costly than false positives?

Required work

- Classical pipeline: HSV colour histograms and a Gaussian mixture model per class.
- Decision rule: Bayesian decision theory with asymmetric loss.
- Modern pipeline: frozen EfficientNet-B0 with a trained classification head.
- Uncertainty: use Monte Carlo Dropout or an equivalent uncertainty estimate.

Expected outputs: ROC curves, reliability or calibration analysis, uncertainty examples, and a comparison of classical and deep triage decisions.

Theory focus: Bayes decision boundaries, asymmetric costs, calibration, and the meaning of MC Dropout.

Topic C – Scale-Space and Invariant Detection: Matching Across Viewpoints

Dataset	VGG Affine Covariant Regions Benchmark: image sequences with known homographies.
Core task	Match local features under increasing viewpoint change.
Main question	Which parts of a local-feature pipeline fail first: detection, description, or matching?

Required work

- Classical detector: Difference-of-Gaussian keypoints from scale-space.
- Classical descriptor: SIFT descriptors and standard matching.
- Modern descriptor: SuperPoint descriptors on the same image pairs.
- Evaluation: repeatability and matching score using the benchmark homographies.

Expected outputs: repeatability and matching-score tables or curves, qualitative match visualizations, and failure cases for difficult viewpoint changes.

Theory focus: scale-normalized LoG, DoG approximation, scale-space axioms, and SIFT orientation assignment.

Topic D – Self-Supervised Representation Learning: STL-10 Linear Evaluation

Dataset	STL-10: labelled and unlabelled natural images in 10 classes.
Core task	Evaluate how much labelled data can be saved by using a pretrained representation.
Main question	Does self-supervised pretraining make simple classifiers work better with fewer labels?

Required work

- Classical baseline: HOG features with a linear SVM.
- Modern representation: frozen DINO ViT-S/16 features.
- Evaluation: train the same linear SVM on different fractions of labelled data.
- Optional comparison: k -NN evaluation on frozen features.

Expected outputs: label-efficiency learning curves, a direct comparison between HOG and DINO, and a discussion of what linear evaluation measures.

Theory focus: DINO self-distillation, representation collapse, linear evaluation, and HOG robustness.

Topic E – Video Understanding: Temporal Action Recognition on UCF-101

Dataset	UCF-101: human-action videos; use the provided 10-class subset.
Core task	Classify short action clips using motion and frame-level appearance.
Main question	When is motion essential, and when is appearance already enough?

Required work

- Classical pipeline: dense optical flow summarized as a motion descriptor.
- Classifier: multi-class SVM on optical-flow features.
- Modern pipeline: frozen ResNet-50 frame features.
- Temporal aggregation: average frame features over 8 sampled frames and train a linear head.

Expected outputs: accuracy, confusion matrix, inference-speed comparison, and examples where temporal order or motion changes the result.

Theory focus: optical-flow constraint, aperture problem, Lucas–Kanade assumptions, two-stream recognition, and temporal pooling limitations.